

Ex. No: 05	IMPLEMENTATION OF NAVIGATION AND ROUTING
Date of Exercise	08.09.2026
Git-hub Host Link	https://urk23cs1203.github.io/Flutter-Lap-Ex5/

Aim:

To implement navigation between multiple Flutter screens using Navigator and named routes.

Description:

This experiment demonstrates how to navigate between multiple screens in a Flutter application. The application contains **Home, Profile, and Details** screens. Named routes are used for navigation, and Navigator.pop() is used to return to the previous screen. A colorful **purple, pink, and sky-blue** theme is used to improve the user interface.

Procedure:

- Step 1: Create a new Flutter project using VS Code or the terminal.
- Step 2: Create separate Dart files for the Home, Profile, and Details screens inside the lib folder.
- Step 3: Design the Home screen with a button to navigate to the Profile screen.
- Step 4: Design the Profile screen with a button to navigate to the Details screen.
- Step 5: Design the Details screen with a Back button to return to the previous screen.
- Step 6: Define named routes for Home, Profile, and Details inside MaterialApp.
- Step 7: Use Navigator.pushNamed() to navigate between the screens.
- Step 8: Use Navigator.pop() to return to the previous screen.
- Step 9: Apply a colorful user interface using purple, pink, and sky-blue colors, gradients, cards, icons, and buttons.
- Step 10: Run the application using flutter run and check the navigation between all screens.
- Step 11: Verify the navigation flow: **Home** → **Profile** → **Details** and back to the previous screens.
- Step 12: Confirm that navigation and routing work correctly in the Flutter application.

Program:**main.dart -**

```
import 'package:flutter/material.dart';
import 'home.dart';
import 'profile.dart';
import 'details.dart';

void main() {
  runApp(const NavigationApp());
}

class NavigationApp extends StatelessWidget
{ const NavigationApp({super.key});

  @override
  Widget build(BuildContext context)
  { return
    MaterialApp( debugShowCheckedMode
    Banner: false, title: 'Navigation App',

    theme: ThemeData(
      colorScheme:
        ColorScheme.fromSeed( seedColor:
        Colors.purple,
      ),
      useMaterial3: true,
    ),

    initialRoute: '/',

    routes: {
      '/': (context) => const HomePage(),
      '/profile': (context) => const ProfilePage(),
      '/details': (context) => const DetailsPage(),
    },
  );
}
```

home.dart -

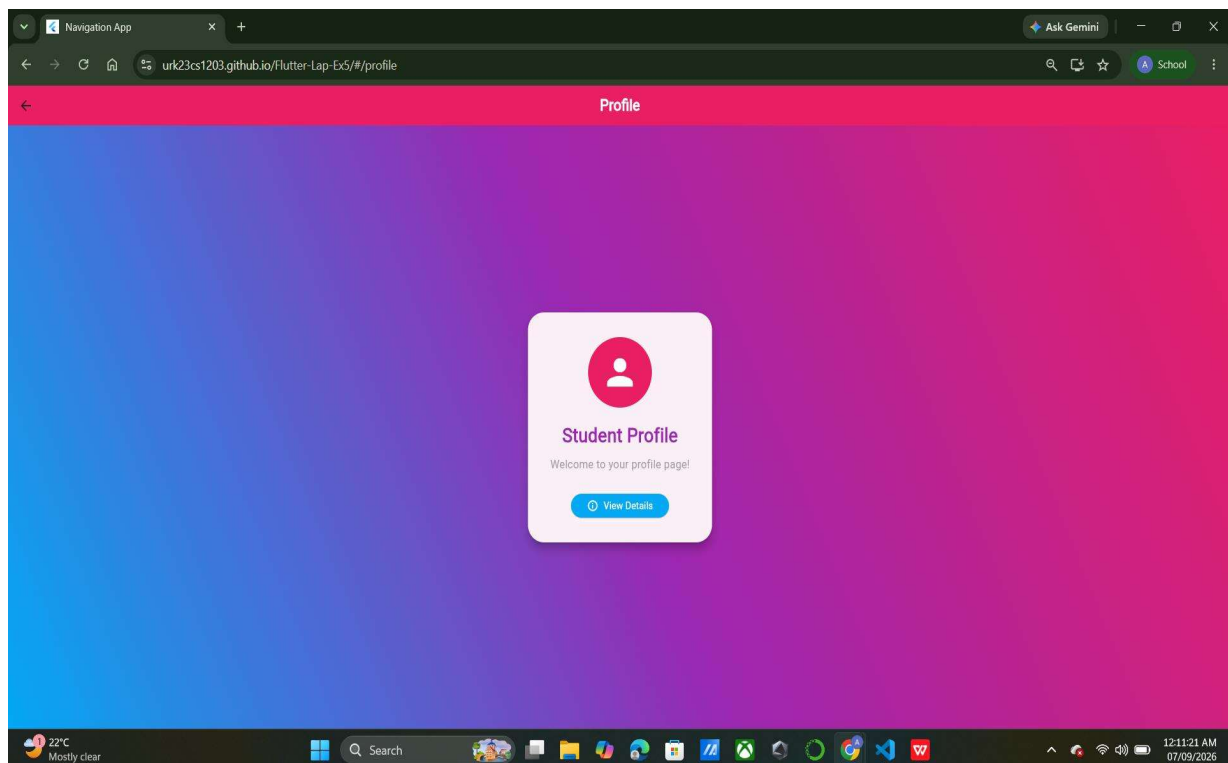
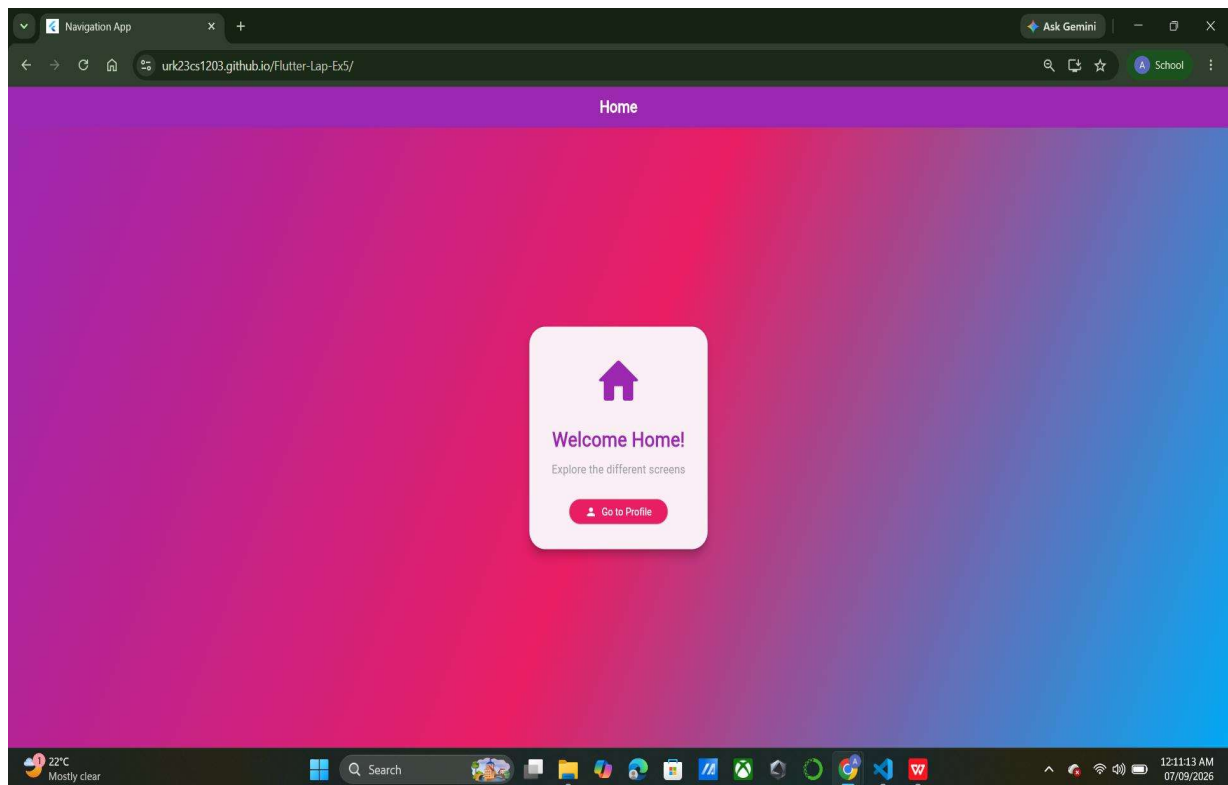
```
import 'package:flutter/material.dart';
class HomePage extends StatelessWidget
{ const HomePage({super.key});
  @override
  Widget build(BuildContext context)
  { return Scaffold(
    appBar:
      AppBar( title:
        const
        Text( 'Home',
          style:
            TextStyle( color:
              Colors.white,
              fontWeight: FontWeight.bold,
            ),
          ),
        centerTitle: true,
        backgroundColor: Colors.purple,
      ),
    body: Container(
      decoration: const
      BoxDecoration( gradient:
        LinearGradient( colors: [
          Colors.purple,
          Colors.pink,
          Colors.lightBlue,
        ],
        begin: Alignment.topLeft,
        end: Alignment.bottomRight,
      ),
    ),
    child: Center(
      child: Card(elevation: 10,
        margin: const EdgeInsets.all(30),
        shape: RoundedRectangleBorder(
          borderRadius: BorderRadius.circular(25),
        ),
      ),
      child: Padding(
        padding: const EdgeInsets.all(35),
        child: Column(
          mainAxisAlignment: MainAxisAlignment.min,
          children: [
            const
            Icon( Icons.home_rounded, size: 80,
```

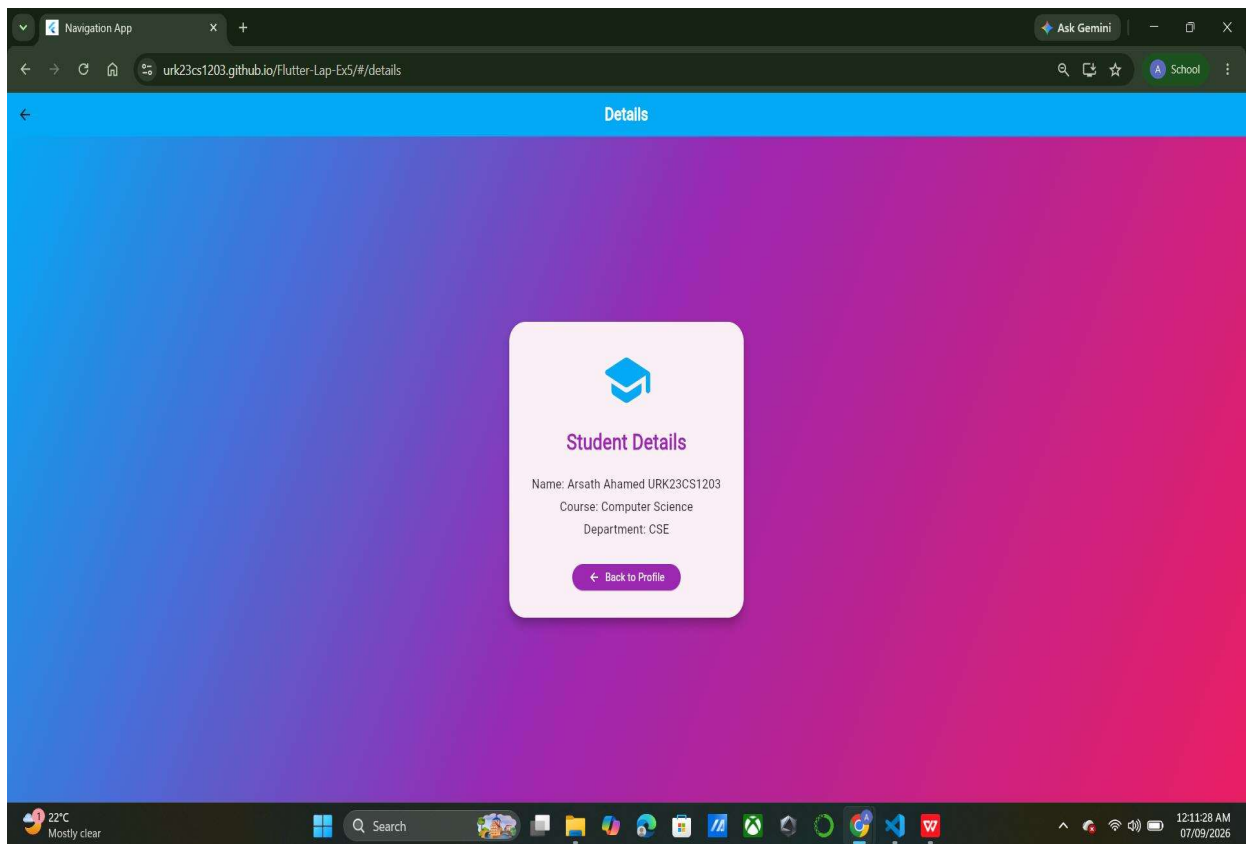

profile.dart -

```
import 'package:flutter/material.dart';
class ProfilePage extends StatelessWidget
{ const ProfilePage({super.key});
  @override
  Widget build(BuildContext context)
  { return Scaffold(
    appBar:
      AppBar( title:
        const
        Text( 'Profile',
          style:
            TextStyle( color:
              Colors.white,
              fontWeight: FontWeight.bold,
            ),
          ),
        centerTitle: true,
        backgroundColor: Colors.pink,
      ),
    body: Container(
      decoration: const
      BoxDecoration( gradient:
        LinearGradient( colors: [
          Colors.pink,
          Colors.purple,
          Colors.lightBlue,
        ],
        begin: Alignment.topRight,
        end: Alignment.bottomLeft,
      ),
    ),
    child:
      Center( child:
        Card( elevation:
          10,
          margin: const EdgeInsets.all(30),
          shape: RoundedRectangleBorder(
            borderRadius: BorderRadius.circular(25),
          ),
          child: Padding(
            padding: const EdgeInsets.all(35),
            child: Column(
              mainAxisAlignment: MainAxisAlignment.min,
              children: [
                const
```


details.dart -

```
import 'package:flutter/material.dart';
class DetailsPage extends StatelessWidget
{ const DetailsPage({super.key});
  @override
  Widget build(BuildContext context)
  { return Scaffold(
    appBar:
      AppBar( title:
        const
        Text( 'Details',
          style:
            TextStyle( color:
              Colors.white,
              fontWeight: FontWeight.bold,
            ),
          ),
        centerTitle: true,
        backgroundColor: Colors.lightBlue,
      ),
    body: Container(
      decoration: const
      BoxDecoration( gradient:
        LinearGradient( colors: [
          Colors.lightBlue,
          Colors.purple,
          Colors.pink,
        ],
        begin: Alignment.topLeft,
        end: Alignment.bottomRight,
      ),
    ),
    child:
      Center( child:
        Card( elevation:
          10,
          margin: const EdgeInsets.all(30),
          shape: RoundedRectangleBorder(
            borderRadius: BorderRadius.circular(25),
          ),
          child: Padding(
            padding: const EdgeInsets.all(35),
            child: Column(mainAxisSize: MainAxisSize.min,
              children: [
                const
                Icon( Icons.school_roun
```


Output:

**Result:**

The Flutter application was successfully implemented with navigation between Home, Profile, and Details screens using named routes and back navigation.